1 Linear Functionals



Definition 1.1 (Linear Functional)

Let V be a vector space over F. An element of $\mathcal{L}(V, F)$ is called a *linear functional* on V.



Exercise 1.2

Fix $a_1, ..., a_n \in F$ and define $f : F^n \to F$ by $f(x_1, ..., x_n) \to a_1x_1 + ... + a_nx^n$.

(a) Show that f is a linear functional on F^n and find the matrix of f relative to the standard bases of F^n and F.

(b) Show that every linear functional on F^n is of this form, for some scalars $a_1, ..., a_n$.



Solution

f is clearly a linear functional (it is a linear combination of elements of F). This is essentially the linear map lemma, which we have already proven. This has matrix:

$$\begin{bmatrix} a_1 & a_2 & ... & a_n \end{bmatrix}$$

Every map is of this form since a map f is determined by its action on the basis, and any map f will send $(0, ..., 1, ..., 0) = a_i$ for 1 in the ith position



Exercise 1.3

Let $V = \mathcal{P}(F)$ and fix $t \in F$. Show that the *evaluation map* $\text{ev}_t : V \to F$ defined by $\text{ev}_t(p) = p(t)$ is a linear functional.



Solution

We have $\text{ev}_t(\alpha p + q) = (\alpha p + q)t = \alpha p(t) + q(t) = \alpha \text{ev}_t(p) + \text{ev}_t(q)$, showing linearity.



Exercise 1.4

Fix a closed interval $[a, b]$ of the real line and let $V = \mathcal{C}([a, b])$, the vector space of continuous functions on $[a, b]$. Show that $f : V \to \mathbf{R}$ defined by $f(g) = \int_a^b g(t)\,\mathrm{d}t$ is a linear functional.



Solution

We have $f(\alpha g + h) = \int_a^b (\alpha g + h)(t)\,\mathrm{d}t = \int_a^b \alpha g(t) + h(t)\,\mathrm{d}t = \alpha \int_a^b g(t)\,\mathrm{d}t + \int_a^b h(t)\,\mathrm{d}t = \alpha f(g) + f(h)$, showing linearity.

2 Dual Space

We have seen previously that for any two vector spaces V and W over F, $\mathcal{L}(V, W)$ is a vector space with respect to the pointwise operations

$$(T_1 + T_2)(v) := T_1(v) + T_2(V)$$
$$(\alpha T)(v) := \alpha T(v).$$



Definition 2.1 (Dual space)

Let V be a vector space over F. We define the *dual space* of V, denoted by V^*, to be the vector space $V^* = \mathcal{L}(V, F)$.



Exercise 2.2

Show that if V is finite-dimensional, then $\dim V^* = \dim V$.



Proof:

We know $\dim \mathrm{Hom}V, W)$ for $\dim V = n$, $\dim W = m$ is mn. But, V has $\dim n$ and F has $\dim 1$, so $\mathrm{Hom}V, F)$ has dim $n \cdot 1 = n$. Since the dimensions are the same and all finite-dimensional vector spaces with the same dimensions are isomorphic, we are done.



Recall:



Lemma 2.3 (Linear Map Lemma)

Let V be a finite-dimensional vector space and W a vector space. Suppose $\{v_1, ..., v_n\}$ is a basis of V and $w_1, ..., w_n$ any n vectors in W. Then there exists a unique linear map $T : V \to W$ such that

$$Tv_k = w_k$$

for each $k = 1, ..., n$.

Suppose V is finite-dimensional and let $B = \{v_1, ..., v_n\}$ be a basis of V. By the linear map lemma, there exists a unique linear functional f_1 such that $f_1(v_1) = 1$ and $f_1(v_j) = 0$ for all $j = 2, ..., n$. Similarly, there is a unique linear functional f_2 such that $f_2(v_2) = 1$ and $f_2(v_k) = 0$ for all $j \neq 2$. Thus, for each $i = 1, ..., n$, there exists a unique linear map f_i such that

$$f_i(v_j) = \begin{cases} 1, & \text{if } i = j, \\ 0, & \text{if } i \neq j \end{cases}$$

We can write these more simply as

$$f_i(v_j) = \delta_{ij},$$

where

$$\delta_{ij} := \begin{cases} 1, & \text{if } i = j, \\ 0, & \text{if } i \neq j \end{cases}$$

is called the *Krockener delta*.



Exercise 2.4

For any $v \in V$, we can expand v in the ordered basis $B = (v_1, ..., v_n)$ as $v = \sum_{j=1}^{n} \alpha_j v_j$. Show that $f_i(v) = \alpha_i$, where f_i is the linear functional defined above. That is, the linear functional f_i picks out the coefficient of the ith basis vector v_i in the expansion of v with respect to B. Thus, we may also call f_i the *ith coordinate function for* B.



Exercise 2.5

Prove that the set of n linear functionals defined above, $B^* = \{f_1, ..., f_n\}$, is a basis of V^*. This basis is called the *dual basis* of the basis B of V.

Proof:

First, we show these span V^*. Every map $f : V \to F$ is determined uniquely by its action on the basis e_i of V (by the linear map lemma), so we have for $v = \sum_{i=1}^{n} \alpha_i v_i$:

$$f(v) = f(\sum_{i=1}^{n} \alpha_i v_i) = \sum_{i=1}^{n} \alpha_i f(v_i) = \sum i = 1^n f_i(v) f(v_i),$$

since $f_i(v) = \alpha_i$ (choosing the i-th coordinate), and hence:

$$f = \sum_{i=1}^{n} f(v_i) f_i,$$

showing span. Independnece is obvious, since if:

$$(\sum_{i=1}^{n} \alpha_i f_i) = \mathbf{0}$$

where $\mathbf{0}$ is the zero function, applying this onto v_i gives:

$$(\sum_{i=1}^{n} \alpha_i f_i)(v_i) = \alpha_i = 0,$$

showing all the α_i are zero.

Remark 2.6

In exercise 1.2 we showed every linear functional $f : F^n \to F$ is of the form $f(x_1, ..., x_n) = a_1 x_1 + ... + a_n x_n$ where $a_i = f(e_i)$. The result of the previous exercise generalizes this result. If V is a finite-dimensional vector space with $\dim V = n$, choose an ordered basis $B = (v_1, ..., v_n)$ and let $B^* = (f_1, ..., f_n)$ be the dual basis of V^*. Then any linear functional $f \in V^*$ can be written uniquely as

$$f = \sum_{i=1}^{n} f(v_i) f_i.$$

If $[v] = (x_1, ..., x_n)$ is the coordinate vector of v relative to B and we set $a_i = f(v_i)$, then:

$$f(v) = \sum_{i=1}^{n} a_i f_i \left(\sum_{j=1}^{n} x_j v_j \right) = \sum_{i=1}^{n} \sum_{j=1}^{n} a_i x_j f_i(v_j) = \sum_{i=1}^{n} \sum_{j=1}^{n} a_i x_j \delta_{ij} = \sum_{i=1}^{n} a_i x_i.$$

Exercise 2.7

Let $B = \{v_1, v_2, v_3\}$ be the basis of $\mathbf{R}^3$ defined by

$$v_1 = (1, 0, -1), v_2 = (1, 1, 1), v_3 = (2, 2, 0).$$

Find the dual basis of $(\mathbf{R}^3)^*$.

Solution

These are linear maps, so we can find each of them by augmenting three columns (one for each map) with the rows being the basis vecotrs on the rows such that they annihilate two vectors and

identity a third. So, we reduce:

$$\left[\begin{array}{ccc|ccc} 1 & 0 & -1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 1 & 0 \\ 2 & 2 & 0 & 0 & 0 & 1 \end{array}\right]$$

which gives:

$$\begin{pmatrix} 1 & 0 & 0 & 1 & 1 & -\frac{1}{2} \\ 0 & 1 & 0 & -1 & -1 & 1 \\ 0 & 0 & 1 & 0 & 1 & -\frac{1}{2} \end{pmatrix}$$

so the dual basis is given by:

$$v_1^* = e_1^* + e_2^* - \frac{1}{2}e_3^*$$
$$v_2^* = -e_1^* - e_2^* + e_3^*$$
$$V_3^* = 0 + e_2^* - \frac{1}{2}e_3^*.$$

3 Trace

An important linear functional on the space of $n \times n$ matrices is the *trace*.

Definition 3.1 (Trace of a matrix)

Define the *trace* of an $n \times n$ matrix A, denoted by $\mathrm{Tr}(A)$, to be the sum of the diagonal elements of A:

$$\mathrm{Tr}(A) = \sum_{i=1}^{n} A_{ii}$$

Example 3.2

Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$. Then $\mathrm{Tr}(A) = 1 + 5 + 9 = 15$.

Exercise 3.3

Calculate $\mathrm{Tr}(I)$, where I is the $n \times n$ identity matrix.

Solution

$$\mathrm{Tr}\left(\begin{bmatrix} 1 & 0 & ... & 0 \\ 0 & 1 & .... & 0 \\ \vdots & \vdots & ... & \vdots \\ 0 & 0 & .... & 1 \end{bmatrix}\right) = 1 + 1 + ... + 1 = n.$$

Exercise 3.4

Show that $\mathrm{Tr} : M^{n\times n}(F) \to F$ is a linear functional.

Proof:

If A, B are matrices, the ijth entry of $\alpha A + B$ is $\alpha A_{ij} + B_{ij}$, at which point the result follows in each entry since the trace is the sum of the diagonal entries $(\alpha A + B)_{ii}$.

Proposition 3.5 (Properties of the trace)

Let A, B be $n \times n$ matrices and c a scalar. Then

(i) $\mathrm{Tr}(A + B) = \mathrm{Tr}(A) + \mathrm{Tr}(B)$,

(ii) $\mathrm{Tr}(cA) = c\,\mathrm{Tr}(A)$,

(iii) $\mathrm{Tr}\left(A^T\right) = \mathrm{Tr}(A)$,

(iv) $\mathrm{Tr}(AB) = \mathrm{Tr}(BA)$.

Exercise 3.6

Prove Theorem 3.5.

Proof:

The first two are obvious by linearity. Since A is square, A^T preserves the diagonal ($A_{ii} \to A_{ii}$), so the third property is obvious. The fourth property is less obvious. We have:

$$\begin{aligned}\mathrm{Tr}(AB) &= (AB)_{11} + (AB)_{22} + ... + (AB)_{nn} \\ &= A_{11}B_{11} + A_{12}B_{21} + ... + A_{1n}B_{n1} \\ &= A_{21}B_{12} + A_{22}B_{22} + ... + A_{2n}B_{n2} \\ &= \vdots \\ &= A_{n1}B_{1n} + A_{n2}B_{2n} + ... + A_{2n}B_{n2}.\end{aligned}$$

Viewing this sum in the columns instead gives BA, since $A_{11}B_{11} + A_{21}B_{12} + ... + A_{n1}B_{1n} = (BA)_{11}$, and so we recover $\mathrm{Tr}(BA)$.

Remark 3.7

It is straightforward to generalize property (iv) of Proposition 3.5 to any finite product of $n \times n$ matrices. For example, for three $n \times n$ matrices, property (iv) implies:

$$\mathrm{Tr}(ABC) = \mathrm{Tr}(BCA) = \mathrm{Tr}(CAB).$$

This is called the *cyclic property* of the trace.

Exercise 3.8

Prove that the trace of an $n \times n$ matrix is a similarity invariant. That is, if A and B are similar $n \times n$ matrices, then they have the same trace.



Proof:

We have $A = CBC^{-1}$, so:

$$\mathrm{Tr}(A) = \mathrm{Tr}\big(CBC^{-1}\big) = \mathrm{Tr}\big(BCC^{-1}\big) = \mathrm{Tr}(BI) = \mathrm{Tr}(B),$$

as desired (using the cyclic property).





Exercise 3.9

Prove that there do not exist $A, B \in M^{n\times n}(F)$ such that $AB - BA = I$.



Proof:

We have:

$$\mathrm{Tr}(AB - BA) = \mathrm{Tr}(AB) - \mathrm{Tr}(BA) = 0 \neq n$$

